SBE16plusV2 SeaCAT Moored

Instrument Configuration

Instrument Serial Number:	16-50677
Instrument Firmware Version:	3.2.1
Zero Conductivity Frequency:	2679.15
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	12099447	600m(600 dBar)

Maximum Depth: 600m

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 50677
CALIBRATION DATE: 25-Jan-26

SBE 16plus V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

a0 = 1.225208e-03
a1 = 2.810056e-04
a2 = -1.635214e-06
a3 = 1.997486e-07

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	559286.636	1.0000	0.0000
4.5000	494047.400	4.5000	-0.0000
15.0000	334568.200	15.0001	0.0001
18.5000	292328.300	18.5000	-0.0000
24.0000	235364.100	23.9999	-0.0001
29.0000	192290.900	29.0001	0.0001
32.5000	166393.500	32.5000	-0.0000

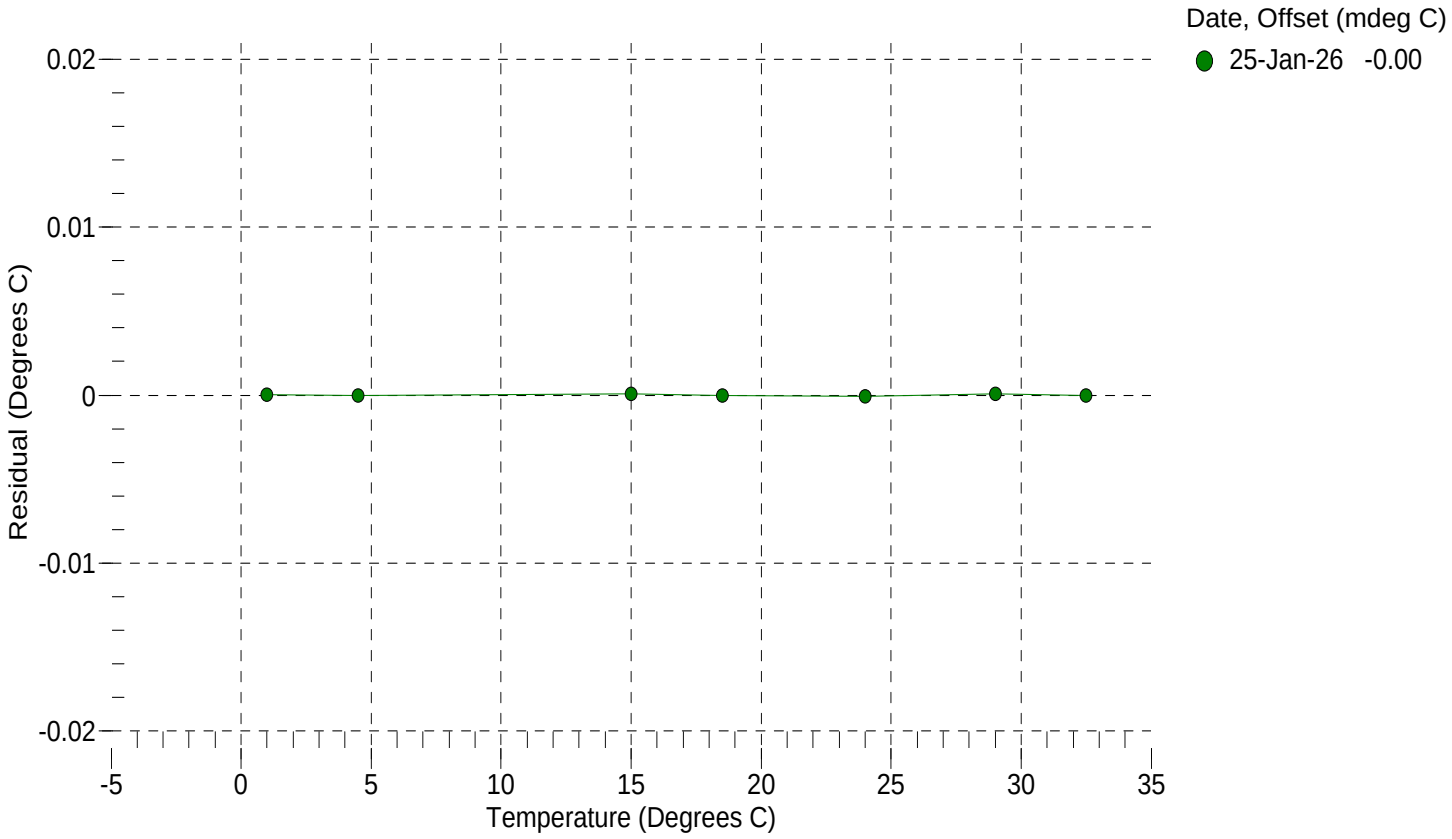
n = Instrument Output (counts)

MV = (n - 524288) / 1.6e+007

R = (MV * 2.900e+009 + 1.024e+008) / (2.048e+004 - MV * 2.0e+005)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(R)] + a2[ln^2(R)] + a3[ln^3(R)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 50677
CALIBRATION DATE: 25-Jan-26

SBE 16plus V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g =  -9.848574e-01
h =   1.374100e-01
i =  -1.650449e-04
j =   3.348161e-05

CPcor = -9.5700e-008
CTcor =  3.2500e-006
```

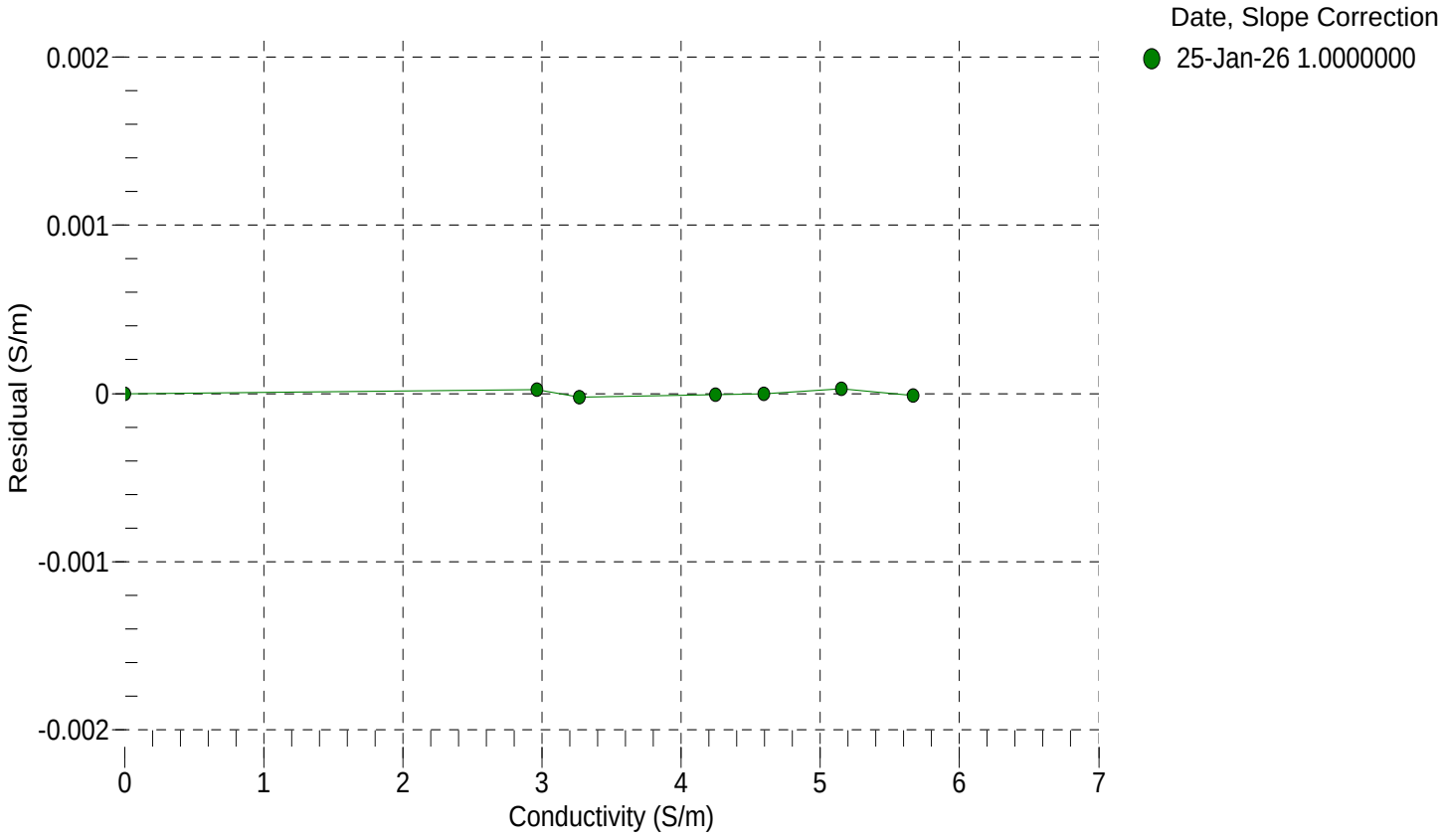
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2679.15	0.0000	0.00000
1.0000	34.6741	2.96497	5359.93	2.9650	0.00002
4.5000	34.6543	3.27094	5562.84	3.2709	-0.00002
15.0000	34.6109	4.24907	6166.09	4.2491	-0.00001
18.5000	34.6016	4.59295	6364.30	4.5929	-0.00000
24.0000	34.5911	5.14882	6671.94	5.1488	0.00003
29.0000	34.5845	5.66861	6946.91	5.6686	-0.00001
32.5000	34.5791	6.03929	7136.37	6.0393	0.00005

f = Instrument Output (Hz) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 50677
CALIBRATION DATE: 23-Jan-26

SBE 16plus V2 PRESSURE CALIBRATION DATA
870 psia S/N 12099447

COEFFICIENTS:

PA0 =	-1.781662e-01	PTCA0 =	5.244466e+05
PA1 =	2.637681e-03	PTCA1 =	-3.977859e+00
PA2 =	2.530939e-11	PTCA2 =	4.293294e-02
PTEMPA0 =	-6.223889e+01	PTCB0 =	2.509625e+01
PTEMPA1 =	5.292632e+01	PTCB1 =	5.000000e-05
PTEMPA2 =	-2.620123e-01	PTCB2 =	0.000000e+00

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.65	530011.0	1.6	14.68	0.00
179.96	592608.0	1.6	179.90	-0.01
359.97	660720.0	1.6	359.90	-0.01
539.98	728751.0	1.6	539.92	-0.01
719.99	796693.0	1.6	719.94	-0.01
869.98	853232.0	1.6	869.93	-0.01
719.90	796723.0	1.6	720.02	0.01
539.92	728777.0	1.6	539.99	0.01
359.91	660739.0	1.6	359.95	0.00
179.91	592624.0	1.6	179.94	0.00
14.65	530006.0	1.6	14.66	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1.81	530104.30
29.00	1.74	530113.80
24.00	1.64	530122.60
18.50	1.54	530131.00
15.00	1.47	530138.10
4.50	1.27	530174.70
1.00	1.20	530186.73

TEMPERATURE (°C)	SPAN
-5.00	25.10
35.00	25.10

y = thermistor output (counts)

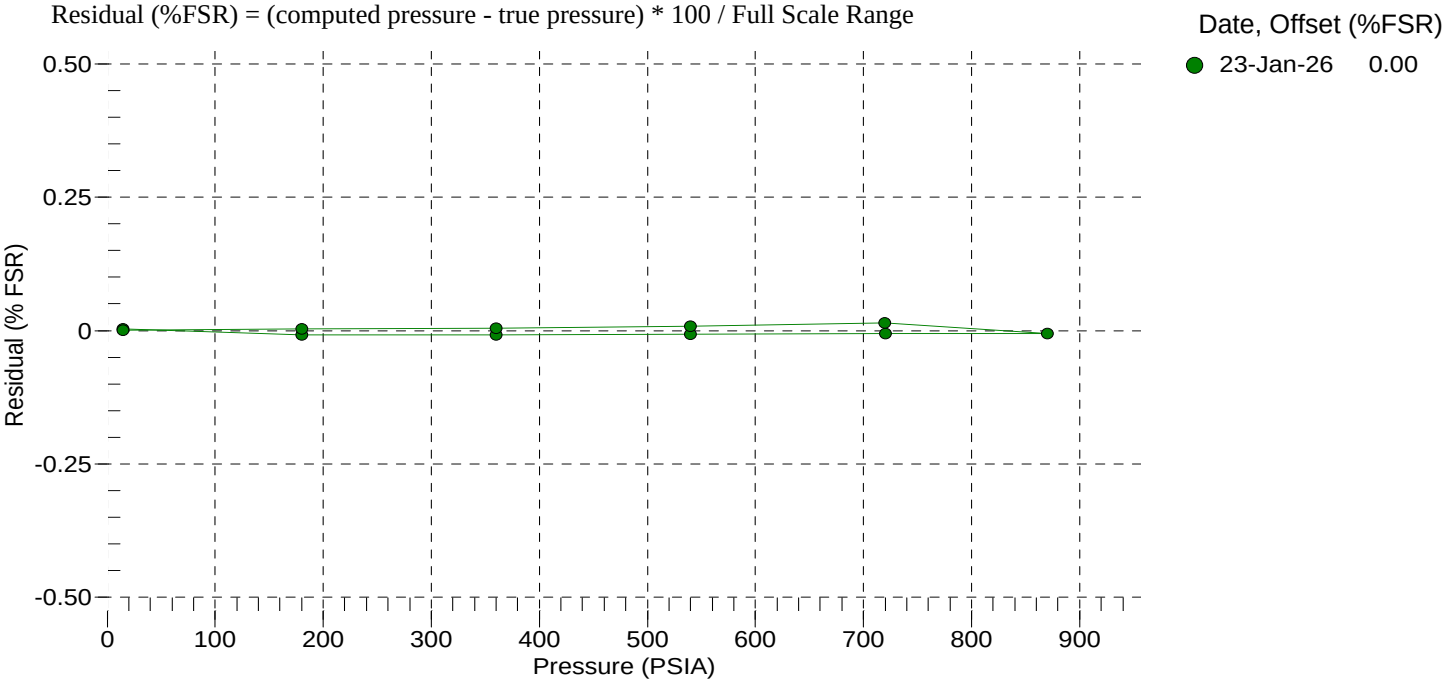
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2025-12-16

Description: SBE-16P SeaCat

Sensor Information:

Model Number: SBE-16P

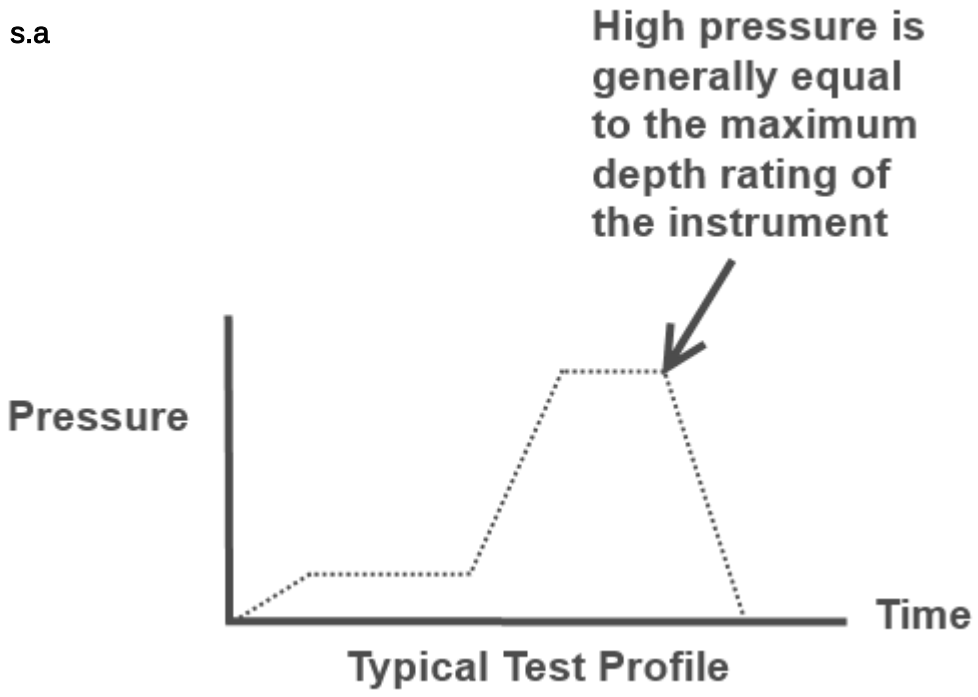
Serial Number: 50677

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	870	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: s.a



Typical Test Profile